Anorectal, Colonic and Antroduodenal Manometry in Children, with Gastric Emptying and Colonic Transit Studies.

Picture three children in a pediatric motility clinic on the same morning. The first is a three-year-old boy who has been constipated since the newborn period. An anorectal manometry study was done last month, awake, and the report says that the rectoanal inhibitory reflex was absent. His parents have been told he may have Hirschsprung disease. The second is a twelve-year-old girl with constipation that has not responded to two years of laxatives. Her high-resolution anorectal manometry report says "dyssynergic defecation," and she has been referred for biofeedback. The third is a nine-year-old boy whose surgeons are planning a cecostomy for antegrade enemas. His colonic manometry, done on the same day as his anesthesia for catheter placement, showed no high-amplitude propagating contractions, and the surgeons are now discussing a colectomy instead.

Each of these reports could be right. Each could also be wrong, and wrong in a predictable way. The clinical question of this chapter is how to use the motility tests of the lower and upper gut in children: anorectal manometry, colonic manometry, antroduodenal manometry, gastric emptying scintigraphy, and radiopaque marker studies of colonic transit. The main idea is that each test answers one narrow question, that none has robust pediatric normal values across every age, and that anesthesia and fecal loading distort most of them. So the first step is always to name the decision the test will change. The stakes are concrete. Missing an absent reflex delays the diagnosis of Hirschsprung disease. Over-reading adult criteria labels healthy children as dyssynergic. And a colonic study read on the day of anesthesia can send a child to the wrong operation.

We will move through the topic in order: the anorectal reflex and what an absent reflex means, the evidence for manometry against biopsy in Hirschsprung disease, the effect of anesthesia, the problem of dyssynergia, the child who is still obstructed after a pull-through operation, colonic manometry and who needs it, antroduodenal manometry and pseudo-obstruction, gastric emptying, and colonic transit. We finish with the guidelines, the traps, and what to do differently in clinic.

A note on sources. The North American consensus documents are the 2017 anorectal and colonic manometry consensus by Rodriguez and colleagues for the American Neurogastroenterology and Motility Society and the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as ANMS and NASPGHAN; the 2018 esophageal and antroduodenal manometry consensus by Rosen and colleagues from the same two societies; and the 2025 NASPGHAN position paper on refractory constipation by Kilgore and colleagues. From Europe we use the 2018 recommendations on pediatric intestinal pseudo-obstruction led by the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN, with Thapar as first author; the 2022 ESPGHAN motility working group review of colonic function tests by Vandenplas and colleagues; and the 2020 Hirschsprung disease guideline of the European Reference Network for rare inherited and congenital anomalies, known as ERNICA, by Kyrklund

and colleagues. For gastric emptying we use the 2008 consensus from the American motility and nuclear medicine societies by Abell and colleagues, the pediatric normal values from Boston by Ng and colleagues in 2020, and the 2025 joint pediatric guideline from the American and European nuclear medicine societies by Lai and colleagues.

Part one. The rectoanal inhibitory reflex.

The rectoanal inhibitory reflex, usually shortened to RAIR, is the reason anorectal manometry matters in Hirschsprung disease. It is an intramural reflex, meaning it runs entirely through the nerves inside the bowel wall. Here is the sequence. A balloon in the rectum is inflated in steps; the international protocol starts at thirty milliliters or more. Stretch receptors in the rectal wall are detected by the myenteric neurons. Inhibitory neurons, which use nitric oxide, then relax the internal anal sphincter, the smooth-muscle ring that provides most of resting anal tone. The manometry catheter sees this as a drop in sphincter pressure. The 2025 NASPGHAN paper defines a RAIR as a fall in internal sphincter pressure of fifteen percent or more, with a return to baseline. If there are no ganglion cells, the reflex cannot happen. So an absent RAIR means one of three things: aganglionosis, a condition called internal anal sphincter achalasia in which ganglia are present but the sphincter still does not relax, or a technical miss.

The reflex is present from birth, including in preterm infants. In a Chinese study by Tang and colleagues of one hundred and eighty healthy newborns, eighty-five of them preterm, a RAIR was elicited in every single baby. In a small Dutch study by de Lorijn and colleagues of ten preterm infants of twenty-eight to thirty-one weeks with delayed passage of meconium, a normal RAIR was present in all ten. So delayed meconium in a preterm infant is not, by itself, a sign of aganglionosis.

The international anorectal physiology working group's London classification, published in 2020 for adults, offers a useful order for reading any study. First the RAIR, then anal tone and contractility, then rectoanal coordination, then rectal sensation. An absent RAIR is classed as a major finding. Dyssynergia, by contrast, is classed only as a minor or inconclusive finding, because the numbers overlap so much with healthy volunteers. We will come back to that.

Part two. Hirschsprung disease: manometry screens, biopsy decides.

How good is manometry at finding Hirschsprung disease? The best summary is a systematic review by de Lorijn and colleagues from 2006, which pooled twenty-four studies. Rectal suction biopsy had a sensitivity of ninety-three percent and a specificity of ninety-eight percent, in nearly one thousand patients. Anorectal manometry had a sensitivity of ninety-one percent and a specificity of ninety-four percent, in four hundred patients. The contrast enema was clearly worse, with a sensitivity of seventy percent and a specificity of eighty-three percent. If you picture the bar chart, the biopsy and manometry bars are almost the same height, and the enema bars sit well below them. Manometry did not differ significantly from suction biopsy, but histology remains the reference standard. In newborns specifically, high-resolution manometry was compared with barium enema in sixteen suspected cases in the Tang study: sensitivity eighty-nine versus seventy-eight percent, and specificity eighty-three versus only seventeen percent.

So what do you do with an absent RAIR? The ERNICA guideline says to consider rectal biopsy. Its recommendations on the biopsy itself are practical. Take the biopsy at least two centimeters above the dentate line, or about three centimeters from the anal verge, on the posterior or lateral wall. A biopsy taken too low lands in the normal zone of physiological hypoganglionosis just above the sphincter and can be falsely called abnormal. Take two or three suction biopsies, and make sure they include enough submucosa, or they are non-diagnostic. An open biopsy is equally accurate. Ask for calretinin staining, and possibly peripherin, especially in premature infants. In Hirschsprung disease calretinin-positive nerve fibers are absent, and acetylcholinesterase staining is increased. The contrast enema is complementary, useful for mapping the transition zone before surgery, but it never replaces histology.

If the RAIR is absent and the ganglia are normal, think of internal anal sphincter achalasia. The manometry looks the same as Hirschsprung disease but the biopsy is normal. In a series by Doodnath and Puri of twenty-four children treated with internal sphincter myectomy and followed for four to fourteen years, sixty-two and a half percent had normal bowel movements without laxatives, a quarter needed small doses of laxatives, and none became incontinent.

Part three. Anesthesia and the reflex.

Many young children cannot cooperate with an awake study, so anorectal manometry is often done under anesthesia, or the result of an awake study is accepted even though the child was crying. Both choices have a cost. In a study by Baaleman and colleagues in 2023, thirty-four children had anorectal manometry both awake and under general anesthesia. Resting anal pressure fell from about seventy millimeters of mercury awake to about forty-six under anesthesia. In nine of thirty-four children, about a quarter, a RAIR was seen only under anesthesia, which means the awake study alone would have sent them for biopsy. In four of thirty-four, about twelve percent, the study under anesthesia was inconclusive because the resting pressure fell so low that a relaxation could not be shown. The practical rule is that an absent RAIR in an awake, crying child may be artifact, and a low-pressure study under anesthesia may be unreadable. The report should say what state the child was in, and if the result is unclear, the study should be repeated before a biopsy.

Part four. The dyssynergia problem.

Dyssynergic defecation means that the anal sphincter does not relax, or even contracts, when the child pushes. Adult criteria from high-resolution anorectal manometry are often applied to children, and that creates a problem. In a study of three-dimensional high-resolution anorectal manometry in two hundred and five constipated children by Banasiuk and colleagues, adult criteria labeled fifty-three percent of constipated children as dyssynergic, but they also labeled forty-six percent of healthy control children as dyssynergic. The difference was not significant. The measurable difference in anal relaxation during bear-down was small: about ten percent relaxation in constipated children against twenty percent in controls. The same group proposed a pediatric cut-off of thirty-one percent relaxation, but even that labeled about sixty-nine percent of constipated children. And the 2025 NASPGHAN paper cites a randomized trial that found no

added outcome benefit from doing anorectal manometry in children with severe constipation.

Why does this happen? Simulated defecation, lying on one side with a catheter in place and an unfamiliar person watching, is not defecation. Even in adults, the London classification calls dyssynergia only a minor finding. What does matter, according to the 2025 NASPGHAN paper, is anal spasm or prolonged relaxation, which should prompt a spinal evaluation, usually with magnetic resonance imaging of the spine. The same paper says the bear-down maneuver during manometry can substitute for a balloon expulsion test. The lesson is not to treat a label.

Part five. The child who is still obstructed after a pull-through.

After surgery for Hirschsprung disease, some children continue to have obstructive symptoms: distension, constipation, enterocolitis. The American Pediatric Surgical Association's Hirschsprung interest group, in guidelines led by Langer in 2017, lists the causes to consider. There may be a mechanical obstruction such as a stricture or twist; persistent or acquired aganglionosis, or a pull-through of the transition zone; internal sphincter achalasia; dysmotility of the bowel above; or functional megacolon from withholding. The first step, recommended by both the American surgeons and ERNICA, is to exclude the mechanical causes with an examination under anesthesia. Then, if no mechanical cause is found, ERNICA strongly recommends intersphincteric botulinum toxin.

Colonic manometry can sort the remaining children. In a study of forty-six symptomatic children after pull-through published in the American Journal of Gastroenterology in 2000, four patterns emerged. Eighteen children had high-amplitude contractions running through the neorectum all the way to the sphincter, and they tended to soil. Nine had normal manometry, and they tended to withhold stool. Fifteen had absent high-amplitude contractions or simultaneous contractions, and they were constipated. Four had a hypertensive internal sphincter above eighty millimeters of mercury. Treatment guided by these patterns improved bowel frequency in seventy-two percent and pain in eighty percent.

Part six. Colonic manometry.

Colonic manometry records pressure along the colon through a catheter usually placed by colonoscopy. The key event is the high-amplitude propagating contraction, abbreviated HAPC, a contraction of more than seventy-five millimeters of mercury that travels at least ten to thirty centimeters; it is the manometric signature of a mass movement. The ESPGHAN review describes the standard protocol: record for three to six hours, give a meal of at least four hundred kilocalories or twenty kilocalories per kilogram, and then give bisacodyl at zero point two milligrams per kilogram, repeated at zero point four if there is no response. The 2025 NASPGHAN paper describes a normal study as a gastrocolic response after the meal plus HAPCs running from the cecum to the rectosigmoid. There are no published pediatric normal values, so interpretation is based on patterns rather than thresholds.

Bisacodyl matters a great deal. In a high-resolution study by Wessel and colleagues of eighteen children with intractable constipation, only two, about eleven percent, had spontaneous

high-amplitude propagating sequences, but sixteen, about eighty-nine percent, had them after bisacodyl. A study by Gomez and colleagues showed that saline infused into the proximal colon, the way antegrade enemas work, increased colonic contractions but did not reliably trigger HAPCs, whereas bisacodyl did.

The classic patterns were described by Di Lorenzo and colleagues in 1992. Children with functional fecal retention kept their HAPCs, ten of eleven had at least one, and their motility rose after a meal. Children with neuropathy had no HAPCs and no postprandial rise. Children with myopathy had almost no contractions at all.

Who needs colonic manometry? The 2025 NASPGHAN paper is clear. It has no value for guiding medical therapy, and should be reserved until medical treatment is exhausted and surgery is being considered. It should then be used to choose the type and timing of surgery, and before taking down an ostomy. The outcome data support this. In a study by van den Berg and colleagues of thirty-two children with a cecostomy, seventy-eight percent had a successful outcome; absent HAPCs throughout the colon predicted failure, and a normal response to bisacodyl predicted success. Colonic dilatation by itself did not predict dysmotility.

Two practical points come from the ESPGHAN review and the NASPGHAN paper. Record for at least six hours, and if the study is abnormal on the day of anesthesia, repeat it the next day, because in one series nearly half of abnormal studies became normal on the following day. And after a diverting ileostomy, wait at least twelve months before repeating the study.

Part seven. Antroduodenal manometry and pseudo-obstruction.

Antroduodenal manometry records pressure in the gastric antrum and the duodenum, usually through a catheter placed endoscopically or under fluoroscopy. Its main role is in pediatric intestinal pseudo-obstruction, a group of disorders in which the bowel behaves as if it were obstructed but there is no mechanical blockage. The 2018 ESPGHAN-led recommendations require symptoms for two months from birth, or six months if they begin later, plus at least two of four criteria: objective evidence of neuromuscular involvement on manometry, histology or transit; recurrent or persistent dilated bowel loops with air-fluid levels; a genetic or metabolic abnormality definitively linked to pseudo-obstruction; and inability to maintain nutrition or growth on oral feeding. ESPGHAN recommends antroduodenal manometry in all children with presumed pseudo-obstruction, with ninety-two percent agreement among the experts, and states that a normal study conclusively rules it out. It also says that radiopaque markers, scintigraphy, hydrogen breath tests and the wireless capsule are not recommended for measuring small bowel transit in this setting, while gastric emptying belongs in the work-up.

Reading the tracing comes down to a few patterns. A normal study shows the migrating motor complex, abbreviated MMC, during fasting, with its phase three, the burst of regular contractions that sweeps down the small bowel, and a fed pattern after a meal. In a study by Cucchiara and colleagues of twelve children with suspected pseudo-obstruction and normal manometry, the final diagnoses included Munchausen syndrome by proxy, celiac disease, malrotation and Crohn disease. A neuropathic pattern shows disorganized, uncoordinated activity, with absent or

abnormal phase three. A myopathic pattern preserves the organization but the amplitude is low, twenty millimeters of mercury or less. ESPGHAN warns that myopathy should be diagnosed confidently only when there are no dilated loops, because dilated bowel mechanically lowers the measured amplitude. In children dependent on parenteral nutrition, Tomomasa and colleagues described persistent low-amplitude contractions and sustained tonic contractions that they did not see in other groups. And healthy children do not always show a spontaneous phase three. In a study by Uc and colleagues of eighteen asymptomatic children, fourteen had a fasting phase three, and erythromycin induced it in the other four. So a missing spontaneous phase three is not, by itself, a disease.

Manometry also predicts outcome. ESPGHAN states that a duodenal or jejunal MMC predicts tolerance of enteral feeding, a phase three predicts a response to prokinetics, and a myopathic pattern predicts poorer survival. In a London study by Fell and colleagues of fourteen infants with chronic idiopathic intestinal pseudo-obstruction, all seven with low-amplitude phase three and a low motility index had a poor outcome, against two of seven above that threshold. In the larger Great Ormond Street series by Heneyke and colleagues of forty-four children, about half had a poor outcome overall. Fifteen of twenty-two with myopathy did poorly, and thirteen of sixteen with associated malrotation did poorly. Prokinetics helped only six of twenty-two, while an ileostomy relieved symptoms in sixteen of thirty-one.

Part eight. Gastric emptying: four hours, not two.

The standard solid gastric emptying study comes from the 2008 consensus of the American motility and nuclear medicine societies. The meal is egg white, two slices of bread, jam and water, about two hundred and fifty-five kilocalories, labeled with technetium-ninety-nine-m sulfur colloid, with images at zero, one, two and four hours. Delayed emptying is defined as more than ninety percent of the meal retained at one hour, more than sixty percent at two hours, or more than ten percent at four hours. Rapid emptying is less than thirty percent retained at one hour. The 2025 joint pediatric guideline from the American and European nuclear medicine societies adapts this for children: a fast of at least four hours, or three for infants; egg substitute, toast and water, with at least half eaten; cheese as an alternative for egg allergy; and for children under five, a milk or formula study with sixty minutes of dynamic imaging and later static images.

Do the adult limits apply to children? Ng and colleagues analyzed one thousand one hundred and fifty-one pediatric solid-meal studies in children aged six and older. The median retention was seventy-two percent at one hour, thirty-nine percent at two hours, fifteen percent at three hours and six percent at four hours. The ninety-fifth percentiles, the upper limits of normal, were ninety-three, sixty-five, thirty-three and ten percent. If you picture the chart, the pediatric upper limits sit almost exactly on top of the adult cut-offs of ninety, sixty and ten percent. The authors concluded that adult standards can be used in children six and older, and the four-hour limit of ten percent held for cheese meals and for partial meals of at least half.

Why insist on four hours? Edwards and colleagues studied one hundred and nine children and found delayed emptying in about thirteen percent at two hours but about twenty-four percent at

four hours. Fourteen of ninety-five studies that were normal at two hours, about fifteen percent, were abnormal at four hours, and symptoms did not differ between the groups, so the clinic cannot predict who needs the extra two hours. Stopping at two hours roughly halves the yield. Before the test, stop opioids, prokinetics and anticholinergic drugs, record the blood glucose in a child with diabetes, and record how much of the meal was eaten. The carbon-thirteen octanoate breath test is an accepted alternative in the pseudo-obstruction work-up, while the wireless motility capsule has only small pediatric series and is no longer supplied.

Part nine. Colonic transit with radiopaque markers.

Radiopaque markers are the simplest physiological test in pediatric gastroenterology. There are two common protocols. In the Metcalf method, the child swallows twenty markers on each of three days and has an abdominal X-ray on day four; the number of markers times one point two gives the transit time in hours. In the Bouchoucha method, the child swallows ten markers a day for six days and has an X-ray on day seven; the number of markers times two point four gives the transit time, with an upper limit of about sixty-two hours. Normal values in children come from several small studies. In Arhan's 1981 study, mean transit was about eight hours in the right colon, nine in the left colon and twelve in the rectosigmoid. Bautista Casasnovas found about eleven, twelve and fifteen hours, with a total of about thirty-eight hours and an upper limit of about fifty hours. Wagener found about five and a half, six and eight hours. In every series, the rectosigmoid holds markers longest, which is the normal pattern in children.

The distribution matters. More than half the markers in the rectosigmoid suggests outlet obstruction, while markers retained throughout the colon suggest slow transit. But disimpact first, because fecal loading slows transit and fakes slow-transit constipation. In de Lorijn's study of one hundred and sixty-nine constipated children, about half had normal colonic transit, which means that a history-based diagnosis of constipation is often wrong; these children often have functional fecal incontinence without retention, and they need behavioral treatment, not escalation. A total transit above one hundred hours predicted a poorer outcome at one year, while rectal impaction predicted a better one. The 2025 NASPGHAN paper recommends markers for an equivocal history and to decide who needs colonic manometry, noting that no protocol is validated in children. Colonic scintigraphy is an alternative but not routine; in one series of one hundred and one children, the geometric center of the isotope at twenty-four and forty-eight hours was about four and five in normal transit, against about two and a half and three and a half in slow transit.

Part ten. Putting it together, and the guidelines.

For a child with refractory constipation, a sensible sequence uses each test for the decision it changes. First, confirm adherence and clear the rectum. Second, use markers to separate slow transit, outlet obstruction and normal transit. Third, use anorectal manometry to establish whether the RAIR is present, and look for spasm that would prompt spinal imaging. Fourth, if the RAIR is absent, perform suction biopsies with calretinin staining. Fifth, perform colonic manometry only when surgery is being planned. For a child with vomiting and distension, start with imaging to exclude mechanical obstruction and malrotation, then gastric emptying to four hours, then

antroduodenal manometry if pseudo-obstruction is suspected. For every study, record the state of the child, awake, sedated or anesthetized, the drugs taken, and the question asked.

The North American and European documents are largely aligned. Both use anorectal manometry to screen for the RAIR and send an absent reflex to biopsy. Both reserve colonic manometry for surgical decisions and describe the same protocol, with no pediatric normal values. Both endorse antroduodenal manometry in suspected pseudo-obstruction, with ESPGHAN going further and saying a normal study rules it out. Gastric emptying is now covered by a joint American and European pediatric guideline, and the adult limits have been shown to apply from age six. The protocols for radiopaque markers differ: the NASPGHAN paper describes adult protocols with films on day five, or on days four and seven, while the European review describes the Metcalf and Bouchoucha methods. There is a material gap around dyssynergia: adult high-resolution criteria call nearly half of healthy children dyssynergic, and no pediatric threshold is validated. The American College of Gastroenterology and the American Gastroenterological Association offer no pediatric motility guidance.

Part eleven. Traps and pearls.

Four traps catch clinicians repeatedly. The first is accepting an absent RAIR from a crying child or a balloon that was too small; re-test before calling it, and then go to biopsy, not to a contrast enema. The second is acting on a report of dyssynergia; adult criteria over-call, labeling almost half of healthy children, so look for spasm instead. The third is accepting a myopathic antroduodenal tracing in a dilated bowel; dilated loops lower amplitude, so repeat after decompression and confirm with histology. The fourth is accepting a normal gastric emptying study that stopped at two hours; ask for the four-hour image. Two further rules are worth remembering. A colonic study done on the day of anesthesia is abnormal about half the time for the wrong reason, so repeat it the next day before a surgeon acts on it. And transit and colonic studies in a loaded colon measure the feces, not the colon.

The numbers worth carrying into clinic are these. A RAIR is a fall in internal sphincter pressure of fifteen percent or more, present at birth, preterm infants included. For Hirschsprung disease, suction biopsy has a sensitivity of ninety-three and a specificity of ninety-eight percent, manometry ninety-one and ninety-four, and contrast enema seventy and eighty-three. An absent reflex leads to two or three suction biopsies at least two centimeters above the dentate line, with calretinin. Under anesthesia, resting pressure falls from about seventy to about forty-six, and a quarter of children show a reflex that was missed awake. HAPCs appear after bisacodyl in about nine of ten children with intractable constipation. A normal antroduodenal study rules out pseudo-obstruction, and an MMC predicts that the child will feed. And for gastric emptying, more than ten percent retained at four hours is delayed, and a two-hour study misses about half.

What to do differently in clinic.

Before ordering any motility test, write down the decision it will change. If you cannot name one, do not order it. When an anorectal manometry report says the reflex is absent, check whether the child was awake and calm, and what balloon volumes were used; if the answer is unclear, repeat

the study, and if the reflex is truly absent, send the child for suction biopsies with calretinin rather than a contrast enema. When a report says dyssynergia, do not refer for biofeedback on that word alone; look for spasm, and treat the child in front of you. Disimpact before markers, and use markers to find the half of constipated children whose transit is normal. Ask for colonic manometry only when the surgeons are about to operate, insist on bisacodyl and on a next-day recording, and use the presence or absence of propagating contractions to choose the operation. For the child with suspected pseudo-obstruction, exclude malrotation first, then obtain antroduodenal manometry, and let the presence of a migrating motor complex guide how hard you push enteral feeds. And when a teenager with nausea has a normal two-hour gastric emptying study, ask for the four-hour image before telling the family the stomach is normal.

End of chapter.